

WEEKLY INTERNSHIP ACTIVITY 12

Introduction to Textures and Materials

BEFORE

(No Texture / Basic Material)



AFTER

(With Textures & Materials)



What is PBR?

PBR (Physically Based Rendering) is a method used in modern game engines to make materials look realistic by simulating how light behaves in the real world.

Key Components of PBR

1. **Albedo (Base Color):** The actual color of the material (no lighting)
2. **Normal Map:** Adds fake surface detail without extra geometry
3. **Roughness Map**
 - Controls how rough or smooth a surface is
 - Rough = dull, Smooth = shiny
4. **Metallic Map:** Defines if a material behaves like metal or not
5. **Ambient Occlusion (AO):** Adds realistic shadows in small crevices

Why PBR is Important

- Makes objects react to light realistically
- Works consistently in different environments
- Used in engines like Unity and Unreal Engine